

Half Adder:

It adds two 1 bit numbers.

input: X, Y
output: C, S

X	Y	C	S
0	0	0	0
0	1	0	1 ✓
1	0	0	1 ✓
1	1	1 ✓	0

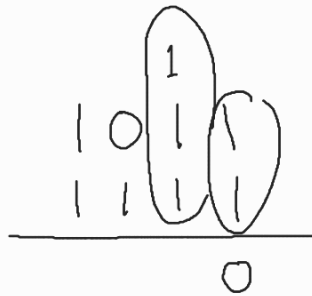
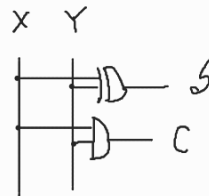
$$\begin{array}{r} 1 \\ + 0 \\ \hline 01 \end{array}$$

$$\begin{array}{r} 1 \\ + 1 \\ \hline 10 \end{array}$$

$$\begin{array}{r} X \\ + Y \\ \hline CS \end{array}$$

$$C = XY$$

$$S = X'Y + XY' = X \oplus Y$$



Full Adder:

It adds three 1 bit numbers.

X	Y	Z	C	S
0	0	0	0	0
0	0	1	0	1 ✓
0	1	0	0	1 ✓
0	1	1	1 ✓	0
1	0	0	0	1 ✓
1	0	1	1 ✓	0
1	1	0	1 ✓	0
1	1	1	1 ✓	1 ✓

$$S = X'Y'Z + X'YZ' + XY'Z' + XYZ$$

$$= X'(Y'Z + YZ') + X(Y'Z' + YZ)$$

$$= X'(Y \oplus Z) + X(Y \odot Z)$$

$$= X'(\underbrace{Y \oplus Z}_A) + X(\underbrace{Y \odot Z}_{A'})$$

$$= X'A + XA'$$

$$= X \oplus A$$

$$S = X \oplus Y \oplus Z$$

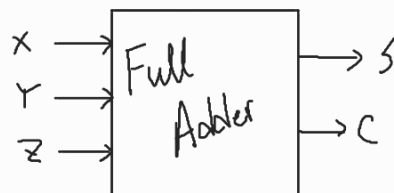
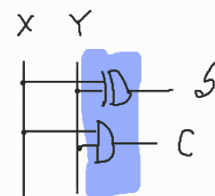
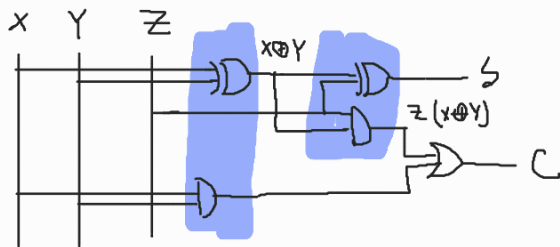
$$C = X'YZ + XY'Z + XYZ' + XYZ$$

$$= Z(X'Y + XY') + XY(Z' + Z)$$

$$C = Z(X \oplus Y) + XY$$

$$S = X \oplus Y \oplus Z$$

$$C = Z(X \oplus Y) + XY$$



1 Full Adder = 2 * Half Adder + OR gate

Circuit Diagram

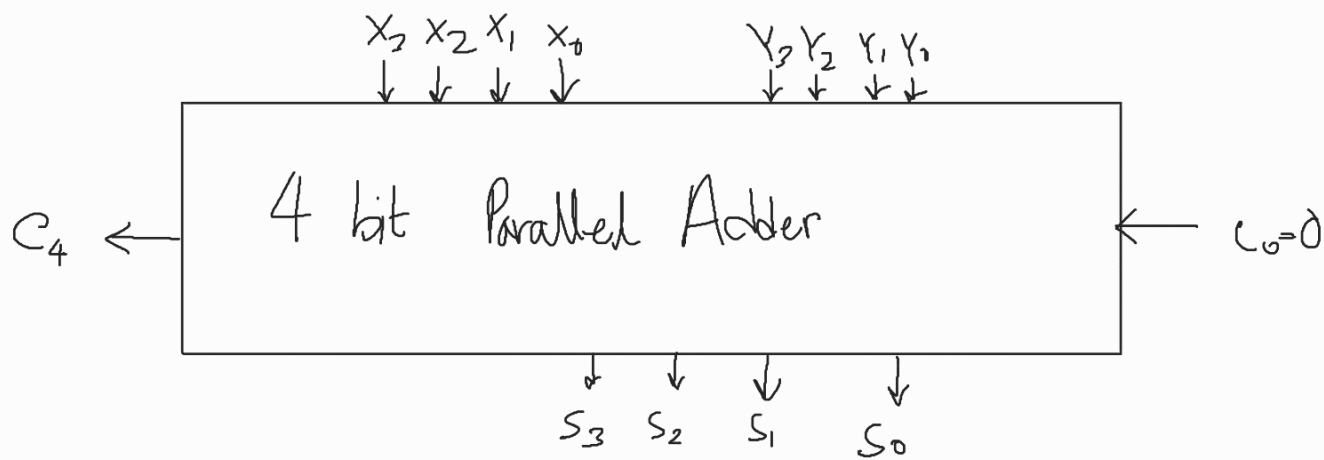
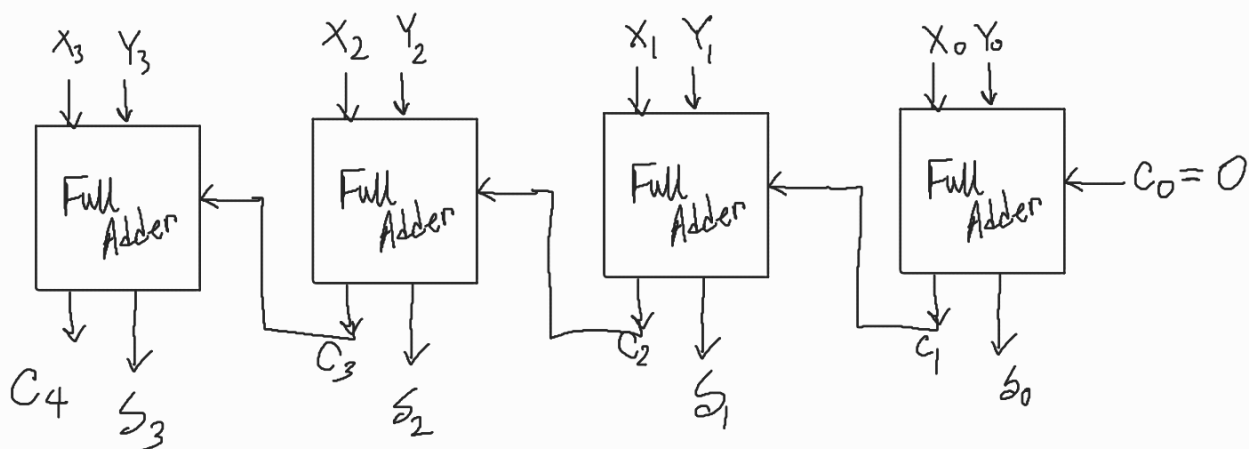
Block Diagram

4 Bit Parallel Adder:

It adds two 4 bit numbers.

$$\begin{array}{r} 1110 \\ X = 1010 \\ Y = 0111 \\ \hline 10001 \end{array}$$

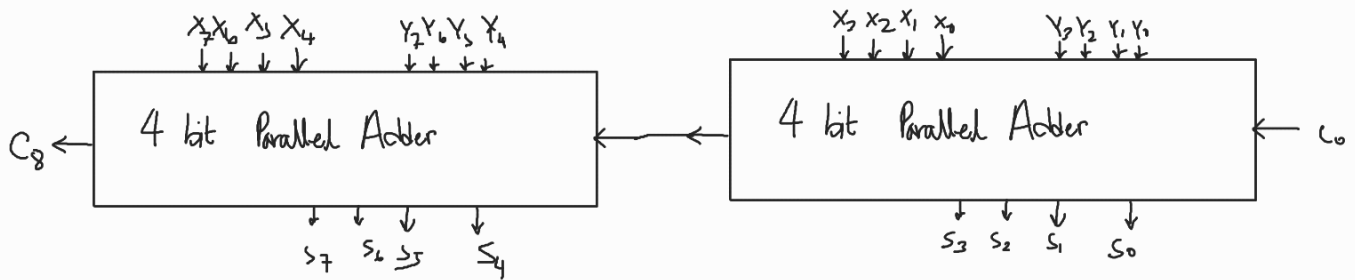
$$\begin{array}{r} C_4 \quad C_3 \quad C_2 \quad C_1 \quad C_0 = 0 \\ X = X_3 \quad X_2 \quad X_1 \quad X_0 \\ Y = Y_3 \quad Y_2 \quad Y_1 \quad Y_0 \\ \hline C_4 \quad S_3 \quad S_2 \quad S_1 \quad S_0 \end{array}$$



$$\begin{array}{r} 10 = X = 1010 \\ 8 = 1000 \\ \hline 18 \quad 10010 \end{array}$$

$$\begin{array}{r} 11 \\ 10 \\ \hline 101 \end{array}$$

2 bit P.A



4 bit parallel Subtractor

$$X - Y = X + (-Y)$$

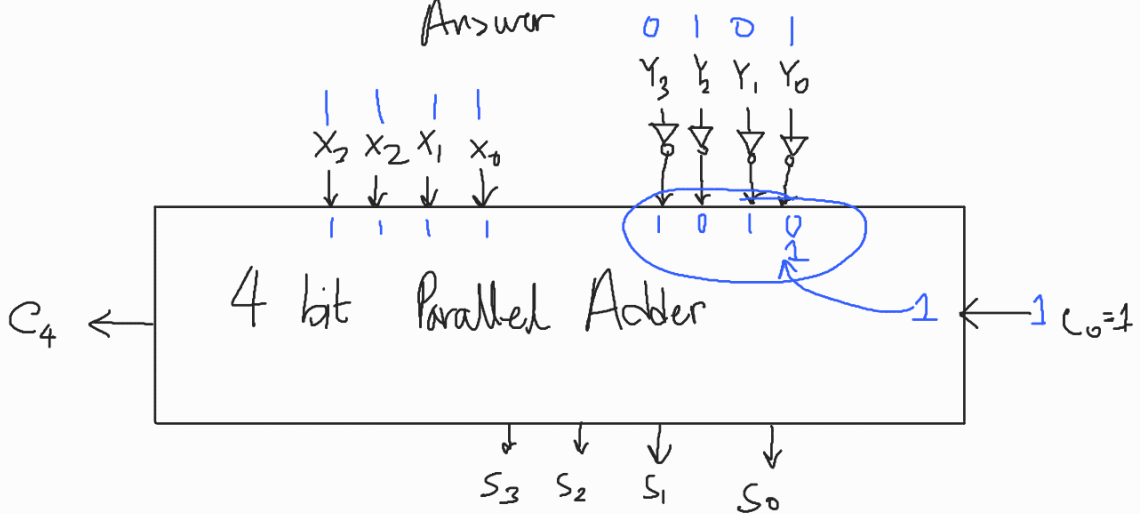
2's

$$\begin{array}{r} 1111 \\ -0101 \\ \hline 1010 \\ +1 \\ \hline 1011 \end{array}$$

Ans

$$\begin{array}{r} 1111 \\ 1010 \\ +10 \\ \hline \end{array}$$

Answer



$$(X - Y)$$

4 Bit Parallel Adder & Subtractor

$$X = 111 = (5)$$

$$Y = 0101 = (5)$$

$$X + Y = 20 = 10100$$

$$X - Y = 10 = 01010$$

XOR

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

